Encoding

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import OneHotEncoder
from sklearn.preprocessing import OrdinalEncoder
```

```
df = pd.read_csv('emp.csv')
```

```
df.head()
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20
4	7654	MARTIN	SALESMAN	7698.0	09/28/2001	1250	1400.0	30

```
df.shape
```

```
(14, 8)
```

```
df2 = df.copy()
```

```
sorted(df2["job"].unique())
```

```
['ANALYST', 'CLERK', 'MANAGER', 'PRESIDENT', 'SALESMAN']
```

```
df2["job"].value_counts()
```

job	count
SALESMAN	4
CLERK	4
MANAGER	3
ANALYST	2
PRESIDENT	1

dtype: int64

```
cols_name = {"job" : {"ANALYST" : 3, "CLERK" : 5, "MANAGER" : 2, "PRESIDENT" :
```

```
cols_name
```

```
{'job': {'ANALYST': 3,
  'CLERK': 5,
  'MANAGER': 2,
  'PRESIDENT': 1,
  'SALESMAN': 4}}
```

```
df2 = df2.replace(cols_name)
df2.head()
```

```
/tmp/ipykernel_23298/1622986362.py:1: FutureWarning: Downcasting behavior in `re
  df2 = df2.replace(cols_name)
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno
0	7839	KING	1	NaN	11/17/2001	5000	NaN	10
1	7698	BLAKE	2	7839.0	05/01/2001	2850	NaN	30
2	7782	CLARK	2	7839.0	06/09/2001	2450	NaN	10
3	7566	JONES	2	7839.0	04/02/2001	2975	NaN	20
4	7654	MARTIN	4	7698.0	09/28/2001	1250	1400.0	30

```
def salCatg(salary):
  if salary >= 3000:
    r=1
  elif salary >= 1500:
    r=2
  else:
```

```
    r=3
  return r
```

```
df2["sal"] = df2["sal"].apply(salCatg)
df2.head()
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno
0	7839	KING	1	NaN	11/17/2001	1	NaN	10
1	7698	BLAKE	2	7839.0	05/01/2001	2	NaN	30
2	7782	CLARK	2	7839.0	06/09/2001	2	NaN	10
3	7566	JONES	2	7839.0	04/02/2001	2	NaN	20
4	7654	MARTIN	4	7698.0	09/28/2001	3	1400.0	30

```
df3 = df.copy()
df.dtypes
```

	0
empno	int64
ename	object
job	object
mgr	float64
hiredate	object
sal	int64
comm	float64
deptno	int64

dtype: object

```
df3["job"] = df3["job"].astype("category")
df3.dtypes
```

	0
empno	int64
ename	object
job	category
mgr	float64
hiredate	object
sal	int64
comm	float64
deptno	int64

dtype: object

```
sorted(df3["job"].unique())
```

```
['ANALYST', 'CLERK', 'MANAGER', 'PRESIDENT', 'SALESMAN']
```

```
df3["job1"] = df3["job"].cat.codes
```

```
df3.head()
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno	job1
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10	3
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30	2
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10	2
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20	2
4	7654	MARTIN	SALESMAN	7698.0	09/28/2001	1250	1400.0	30	4

```
df3["comm"] = df3["comm"].astype("category")
```

```
sorted(df3["comm"].unique())
```

```
[nan, 0.0, 300.0, 500.0, 1400.0]
```

```
df3["comm1"] = df3["comm"].cat.codes
```

```
df3.head()
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno	job1	com
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10	3	
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30	2	
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10	2	
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20	2	
4	7654	MARTIN	SALESMAN	7698.0	09/28/2001	1250	1400.0	30	4	

One Hot Encoding (Dummies)

```
df4 = df.copy()
```

```
sorted(df4["deptno"].unique())
```

```
[np.int64(10), np.int64(20), np.int64(30)]
```

```
df4["depID"] = df4["deptno"]
df4.head()
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno	depID
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10	10
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30	30
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10	10
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20	20
4	7654	MARTIN	SALESMAN	7698.0	09/28/2001	1250	1400.0	30	30

```
pd.get_dummies(df4, columns=["deptno"], prefix=["dno"]).head()
```

	empno	ename	job	mgr	hiredate	sal	comm	depID	dno_10	dn
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10	True	I
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30	False	I
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10	True	I
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20	False	
4	7654	MARTIN	SALESMAN	7698.0	09/28/2001	1250	1400.0	30	False	I

Custom Binary Encoding (Target Encoding)

```
df3["CBE code"] = np.where(df3["hiredate"].str.contains("2001"), 1, 0)
```

```
df3.head()
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno	job1	com
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10	3	
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30	2	
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10	2	
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20	2	

```
df3[["ename", "hiredate", "CBE code"]].head(14)
```

	ename	hiredate	CBE code
0	KING	11/17/2001	1
1	BLAKE	05/01/2001	1
2	CLARK	06/09/2001	1
3	JONES	04/02/2001	1
4	MARTIN	09/28/2001	1
5	ALLEN	02/20/2001	1
6	TURNER	09/08/2001	1
7	JAMES	12/03/2001	1
8	WARD	02/22/2001	1
9	FORD	02/03/2001	1
10	SMITH	12/17/2000	0
11	SCOTT	04/19/2007	0
12	ADAMS	05/23/2007	0
13	MILLER	01/23/2002	0

```
df3[df3["CBE code"] == 1]
```

	empno	ename	job	mgr	hiredate	sal	comm	deptno	job1	co
0	7839	KING	PRESIDENT	NaN	11/17/2001	5000	NaN	10	3	
1	7698	BLAKE	MANAGER	7839.0	05/01/2001	2850	NaN	30	2	
2	7782	CLARK	MANAGER	7839.0	06/09/2001	2450	NaN	10	2	
3	7566	JONES	MANAGER	7839.0	04/02/2001	2975	NaN	20	2	
4	7654	MARTIN	SALESMAN	7698.0	09/28/2001	1250	1400.0	30	4	
5	7499	ALLEN	SALESMAN	7698.0	02/20/2001	1600	300.0	30	4	
6	7844	TURNER	SALESMAN	7698.0	09/08/2001	1500	0.0	30	4	
7	7900	JAMES	CLERK	7698.0	12/03/2001	950	NaN	30	1	
8	7521	WARD	SALESMAN	7698.0	02/22/2001	1250	500.0	30	4	

Ordinal Encoding

```
sorted(df3["deptno"].unique())
```

```
[np.int64(10), np.int64(20), np.int64(30)]
```

```
ord_enc = OrdinalEncoder()
df3["dno_code"] = ord_enc.fit_transform(df3[["deptno"]])
df3[["ename", "deptno", "dno_code"]].head(14)
```

	ename	deptno	dno_code
0	KING	10	0.0
1	BLAKE	30	2.0
2	CLARK	10	0.0
3	JONES	20	1.0
4	MARTIN	30	2.0